

# COM114 Java Programming – Assessment 3

## A Film and TV series Exploration Dashboard

Submission deadline at noon on 17th December 2025.

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This is an individual piece of assessment; you are NOT allowed to work in groups and your code must all be your own work. You must not use code or ideas that are not your own, e.g., taken from the internet or using generative AI tools such as ChatGPT, Gemini or else. Any form of unfair means (e.g. plagiarism or collusion) will be treated as a serious academic offence and will be processed under our University Discipline Regulations. These might result in a loss of marks, a possible failure of the module or even expulsion from our University. Read further details in the section on Academic Integrity in the [UG Student Handbook](#).

You may ask demonstrators for help in understanding the problem, but as this is intended to be an individual assignment, therefore they will not be able to help you directly with programming tasks. In addition to seeking help in practical sessions, you are strongly encouraged to use the Assessment 3 discussion board set up on Blackboard. Posting your questions there will benefit everyone, please, check it regularly. You can subscribe to the discussion board by selecting Follow (see screenshot below).

Discussion Topic



### Learning Outcomes

This assignment will assess your ability to:

- Write a programme from a specification
- Write well-documented code
- Use of classes, objects, inheritance, and polymorphism
- Use of JUnit tests to test a programme
- Use the Java Collections Framework to store and manipulate data
- Use the Java Swing API to build a Graphical User Interface

### Assignment Overview

In this programming assignment, you will create a film and TV series exploration dashboard using Java. The program will start loading a dataset, will store its contents using Java collections and will open a GUI where the user will be able to interact with the dataset to obtain some details about the films and TV series in the dataset. The dashboard will be interactive and will update based on the user selections, and will involve drawing a pie chart using Java2D. Your dashboard will look similar to the one shown in Figure 1.

To get you started and facilitate your programming task, you have been provided with a template starting code that includes several classes, a number of JUnit tests, and the dataset that you will be using.

You need to read this brief **carefully** together with the template code before starting to work on this programming task.

You **must use the starting template** and you **must strictly adhere to the guidance provided**.

This brief is organised as follows:

- [the dataset](#),
- [the template starting code](#),
- [the programming task](#),
- [how to start your programming using the starting template](#),
- [how to submit your code](#),
- [the marking criteria](#), and
- [what to do if you have questions](#).

## The Dataset

You will work on a dataset that has been adapted from an open-source dataset<sup>1</sup> available in Kaggle.

The data file (`titles.csv`) contains plain text and uses the Comma-Separated Values (CSV) file format. The first row provides the column header (i.e. id, title, type, description, runtime, etc), and the remaining rows contain the actual values for each of the properties in the dataset (in the same order as the headers).

The properties for each entry in the dataset are (see CSV file header):

| No. | Header            | Description                                               |
|-----|-------------------|-----------------------------------------------------------|
| 1   | Id                | Unique identifier for each title in the dataset           |
| 2   | title             | Name of the movie or TV show                              |
| 3   | type              | Indicates whether it is a Movie or Show                   |
| 4   | description       | Short summary or plot of the title                        |
| 5   | release_year      | Year the movie or show was released                       |
| 6   | age_certification | Age rating of the title [TV-MA, R, TV-14, PG-13, PG...]   |
| 7   | runtime           | Duration of the movie or show in minutes                  |
| 8   | imdb_id           | Unique identifier for imdb                                |
| 9   | imdb_score        | IMDB rating (scale low [0-4], average (4-7), high (7-10)) |

Table 1. Description of the properties for Movies/shows entries in the dataset.

**Handling of missing values:** If the dataset has missing values (but a correct number of 'columns' or 'fields'), please use a default value, and do not discard the entry. E.g. in the provided dataset, the field 'age\_certification' presents many entries with a missing value. See how in Figure 1 we resolved this situation by having a 'not-specified' option in the pie chart.

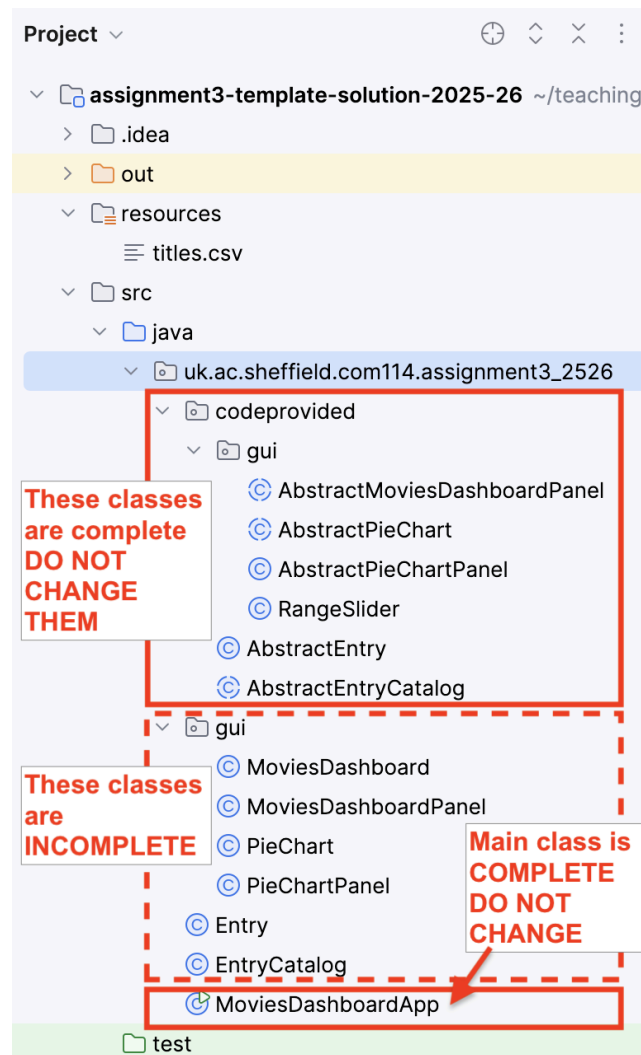
**Handling of incorrect input file:** If the dataset presents entries with too many or too few columns, or provides a string when a number is expected (e.g. 'release\_year') then the execution of the programme should be stopped because the input file is incorrect.

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<sup>1</sup> "IMDB scores dataset" <https://www.kaggle.com/datasets/sufyan145/netflix-movies-and-shows-imdb-scores>.

## The Template Starting Code

You have been provided with a zip file that contains a template project with some starting code. Your template starting project will look like this:



The project is organised in packages according to the purpose of each class:

- The `uk.ac.sheffield.com114.assignment3_2526.codeprovided` package **must not be changed**. It contains essential classes related to your assignment.
- The `uk.ac.sheffield.com114.assignment3_2526.codeprovided.gui` **must not be changed** either. This package contains the classes relevant to the creation of the GUI.
- The `uk.ac.sheffield.com114.assignment3_2526` and `uk.ac.sheffield.com114.assignment3_2526.gui` packages are the ones that contain the classes you need to implement. You have been provided with several basic implementations. **You will need to edit these classes**, including adding imports, comments, new variables, new methods, and removing all the TODO helper comments we have provided.

The main class of this project is `MoviesDashboardApp`, please **do not change this class**, apart from adding JavaDoc comments and removing the TODO comments we provided. This is the class that we will use to run your code during marking.

## The Programming Task

Starting from the template code provided, you will need to:

- **Use** the classes in the **codeprovided packages**.  
As already mentioned in the previous section, you **must not change anything** in these classes, they are meant to help you, and you will need to understand what they do as they are essential for accomplishing this assignment.
- **Complete** the classes in the starting template code that **are not part** of the codeprovided package. The starting implementation compiles but the current implementation does not work as this brief indicates.  
Your tasks are to edit these classes (replacing the basic implementations provided) and appropriately document them.
- **Create new helper classes** to support your implementation. You **should not** place these classes in the codeprovided packages.

Your classes will need:

- to compile,
- be documented. This includes Javadoc for the classes you will edit and the new ones you might need to create,
- use good naming for variables and methods,
- be well formatted, according to the guidance provided in the module,
- remove unused code,
- remove the TODO comments provided as a support for your implementation.

### ***Methods to implement:***

You need to implement all the methods in the `EntryCatalog` class (which extends the `AbstractEntryCatalog` class). These methods will help answer questions of the following type (see in parentheses which method in this class should be used to answer each question):

- What is the total number of entries in the data set?  
(`getNumberOfEntries`)
- What is the earliest release year in the dataset?  
(`getEarliestReleaseYear`)
- What is the average imdb score for all the titles released after 1990?  
(`getAverageImdbScoreTitlesReleasedAfterYear`, note that the provided year should not be included in the average computation)
- What is the average imdb score for all the titles released before 1900?  
(`getAverageImdbScoreTitlesReleasedBeforeYear`, note that the provided year should not be included in the average computation)
- What is the average imdb score for all the titles released between 2000 and 2010?  
(`getAverageImdbScoreTitlesBetweenYears`, note that the provided years should be included in the average computation)

- What is the maximum runtime for all the entries rated G?  
(getMaximumRuntimeRatedAgeCertification)
- What is the minimum runtime for all the entries rated NC\_17?  
(getMinimumRuntimeRatedAgeCertification)
- What is the average runtime of movies rated PG-13?  
(getAverageRuntimeRatedAgeCertification)
- ~~How many~~ Which movies/shows are based on true stories<sup>2</sup>?  
(getTitlesBasedOnTrueStory, note that this method returns a Map object. Using the key-value terminology, the key of the returned map should be either "MOVIE" or "SHOW", and the values, a set with the titles)
- How many titles have an IMDB score greater than 7.0?  
(getNoTitlesImdbScore, note that this method is more general and allows users to provide an operator as one of its parameters)
- What is the average IMDB score of all the TV-Shows?  
(getAverageImdbScoreTitle)
- What is the minimum IMDB score of all the Movies?  
(getMinimumImdbScoreTitle)
- What is the maximum IMDB score of all the TV-Shows?  
(getMaximumImdbScoreTitle)
- ~~How many~~ Which entries are identified as documentaries?<sup>3</sup>  
(getDocumentaries)
- ~~How many~~ Which movies were released in the last five years? Use 2025 as the reference year; see more details on BB Discussion board.  
(getLastFiveYearsMovies)

**Tip:** Some of the above methods will be helpful when completing some of the functionalities required to implement the GUI.

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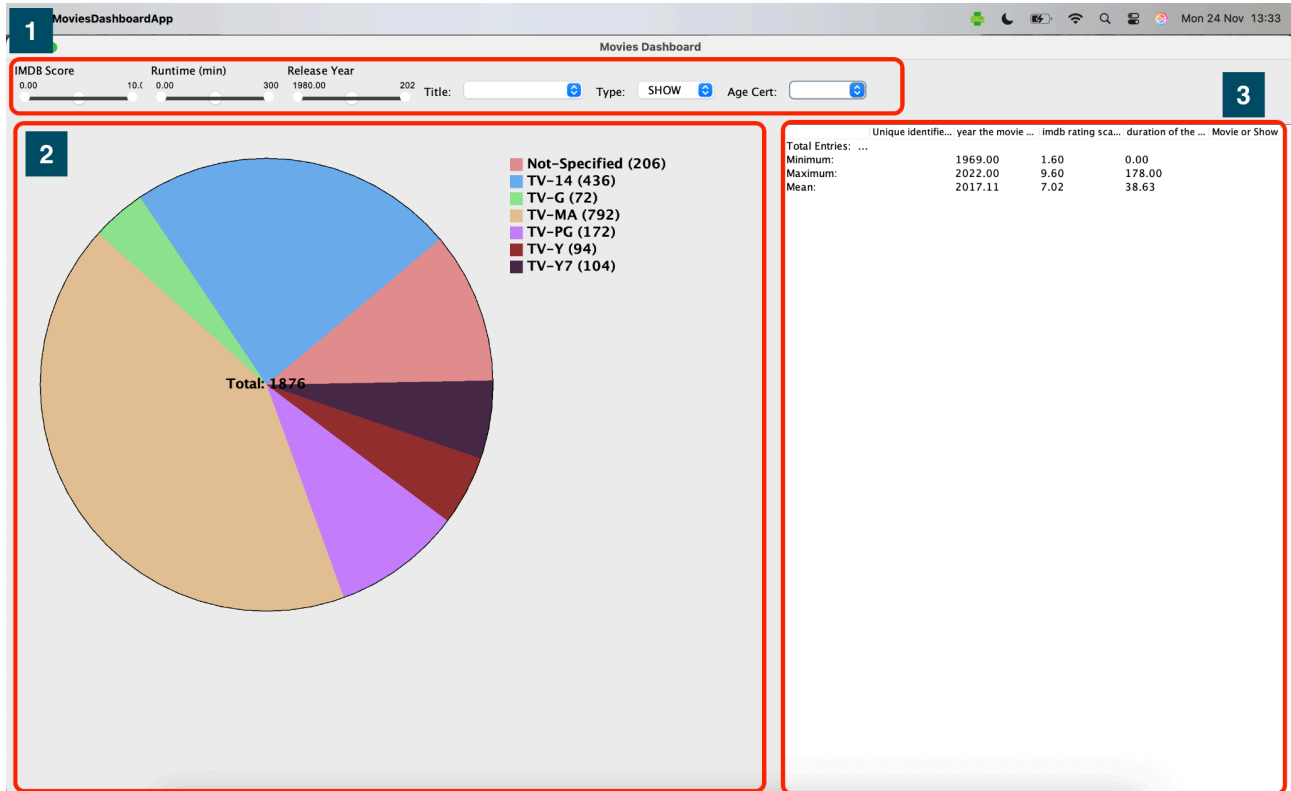
<sup>2</sup> A good way to approach this is to look into the description field in the dataset and search for these relevant keywords: "true story", "true events" and "real events". Then extract the titles from each matched entry.

<sup>3</sup> Focus on identifying the keyword "documentary" in the description field of the dataset and then extract the titles with an exact match.

## The GUI

Figure 1 shows how the Movies Dashboard GUI could look like for the following user selection:

- IMDB Score between 0-10
- Runtime (min) between 0-300
- Release year between 1980-2025
- Type "Show"



**Figure 1. GUI screenshot (for reference only).**

**The Pie Chart shows the total number of entries for the filters specified in Area 1.**

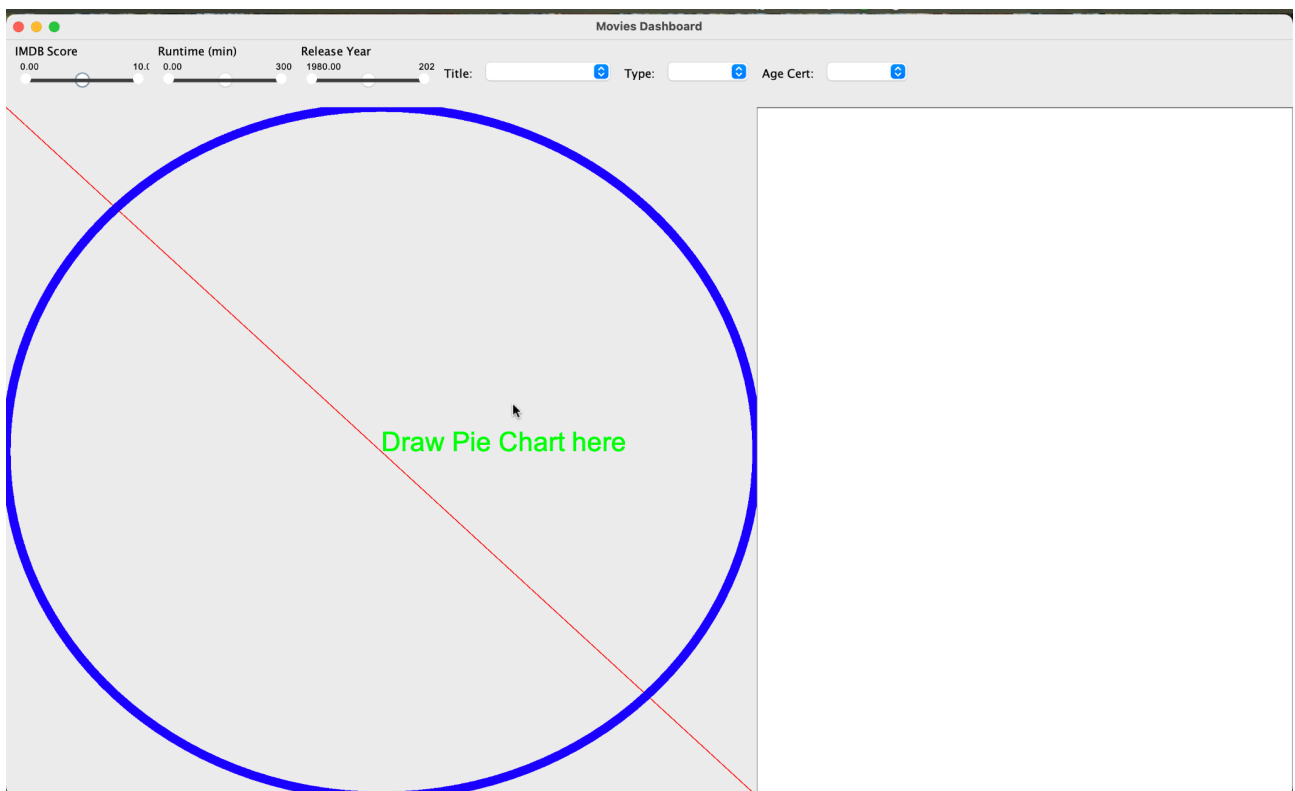
The GUI has 3 main areas with the following functionalities:

- **Area 1 (top):** Allows the user to specify filters that must be applied to the titles' entries in the dataset. This includes a filter using 3 sliders by IMDB Score, Runtime (min) and Release Year. Another way to filter is by using properties like Title, Type and Age Certification. Areas 2 and 3 must be repainted whenever the user changes the filter.
- **Area 2 (left):** Shows a pie chart that displays the distribution of entries by age certification. The chart reflects the current selection after applying filters in Area 1. The class `PieChartPanel` will draw the pie chart. More details are provided in the section 'The Pie Chart'. You are free to innovate on how your pie chart should look like.
- **Area 3 (right):** Shows basic statistics (maximum, minimum and mean) for the resulting dataset after applying the filters specified in Area 1. Only numeric attributes should be shown in this area. You are free to innovate here.

Note that the pie chart has a legend which helps users understand which slice represents which age certification.

Your template starting code provides a basic implementation of the GUI in Figure 1. Please study the class `MoviesDashboardPanel` as it already contains the same three areas as shown in

Figure 1. When you run your template starting code for the first time, your GUI will look like the screenshot in Figure 2.



**Figure 2. GUI screenshot for the template starting code.**  
**The Pie Chart shows a circle, a line and a string. Find that code in your project.**  
**You will need to replace it by a pie chart similar to the one shown in Figure 1.**

### ***The Pie Chart (Area 2)***

To understand the characteristics of the entries in the dataset, the GUI will allow the user to plot a pie chart for the different entry properties. The contents of the pie chart will be updated according to the user selections at the top of the GUI (Area 1).

Pie charts are used in data visualization to represent how a whole set of items is divided into categories. Each segment (or 'slice') of the pie corresponds to a category, and the size of that slice is proportional to the frequency or percentage of items belonging to that category relative to the total.

Please read carefully the classes `AbstractPieChart` and `PieChart`, as the comments inside `AbstractPieChart` provide you detailed guidance about what you need to do.

## How to start this programming assignment

Before you begin programming, you need to:

- Carefully read and understand this brief.
- Understand well the starting template code.
- To complete the classes `Entry`, `EntryCatalog` and `PieChart`, you must have covered the contents up to week 8 (including practical sheets and self-assessment quizzes)
- To complete the classes `MoviesDashboardPanel` and `PieChartPanel`, you must have covered the contents up to week 9.

The main class in this assignment is the class `MoviesDashboardApp`. The first argument it receives is the input file (`titles.csv`) containing the data your assignment will process and represent. This file is in the resources folder in the zip file shared with you.

You **must not change** this class as we will use it to run your code.

```
// do not change this main method
public static void main(String[] args) {
    if (args.length == 0) {
        args = new String[]{
            "./resources/titles.csv"
        };
    }
    MoviesDashboardApp moviesDashboardApp = new MoviesDashboardApp(args[0]);
    moviesDashboardApp.startGUI();
}
```

### ***Classes in the codeprovided package (not to be changed):***

You should use the classes in the codeprovided packages:

`uk.ac.sheffield.com114.assignment3_2526.codeprovided` and  
`uk.ac.sheffield.com114.assignment3_2526.codeprovided.gui`,

**but you must not modify** them. You are expected to familiarise yourself with these classes and the relationship between them. The classes provided are:

- `AbstractMoviesDashboardPanel` is an abstract class that provides a framework for building a GUI panel similar to the one shown in Figure 1. It defines a set of abstract methods that your `MoviesDashboardPanel` will override to provide functionalities described in the handout.
- `AbstractPieChart` is an abstract class that will help represent the pie chart that will be plotted by the class `PieChartPanel`. Your `PieChart` extends this class.
- `AbstractPieChartPanel` is an abstract class that provides the basic code for displaying the `PieChart` object in the GUI. `PieChartPanel` extends this class.
- `RangeSlider` provides a basic implementation of the sliders shown in Area 1 of the GUI.
- `AbstractEntry` is an abstract class providing a very basic implementation of a data entry with a unique identifier.

- `AbstractEntryCatalog` is an abstract class that provides implementation of reading the dataset. It also includes the abstract methods that your `EntryCatalog` will need to override to provide the functionalities described in this handout.

### ***Classes to implement with a basic implementation provided (need to be changed):***

In this project, you have been provided with basic implementations of the classes that you will need to program. They must work together with the classes that have been provided to you. This code needs to be edited as it does not perform the required functionalities. You will need to delete the implementation of the methods and replace it with your own code.

The classes that you can edit are:

- `MoviesDashboardApp`. This is the main class. You will not need to change the code, but you will need to document it and remove the TODO comments provided with the template code.
- `Entry` (this extends `AbstractEntry`).
- `EntryCatalog` (this extends `AbstractEntryCatalog`).
- `PieChart` (this extends `AbstractPieChart`).
- `PieChartPanel` (this extends `AbstractPieChartPanel`).
- `MoviesDashboardPanel` (this extends `AbstractMoviesDashboardPanel`).
- `MoviesDashboard` provides a basic implementation of the main window of your application. You will probably not need to change this class.

### ***Can I add more classes?***

Yes, you can create additional helper classes if needed.

Make sure that you place them in the appropriate package (not in any of the codeprovided packages).

### ***I have found a fantastic library that draws wonderful pie charts, can I use it?***

No, you are meant to use standard Java, in this case Java2D, to produce your pie charts. You cannot use any external package beyond the standard Java API.

### ***Do I need to use the sheffield package?***

No, you do not need to use the sheffield package and you **should not** be using it. In this assignment you are required to use the standard Java API.

### ***How can I use the tests provided to test your solution***

You have been provided with two JUnit test classes in the `src/test` directory of your project.

The provided JUnit tests are meant to help you assess that your solution is 'correct' and that you are working in the right direction. Your submitted code will need to successfully pass all the tests provided. Note that the battery of tests provided doesn't fully test your solution, so even if your solution passes these tests, you need to be sure that your solution is actually fully correct, as during marking, other tests will be run on your code.

## Submission guidance

You **must submit the code** on Blackboard as a **zip file** similar to the one you have been provided with the starting template code. This means that the zip file must:

- respect the same folder structure,
- contain only the source code (the \*.java files), this also includes the code provided package.

Make sure **not to submit**:

- \*.class files (in the out folder).

To clarify this, we will provide a supporting video to show how your code needs to be zipped before submission, ensuring that this step is done correctly.

## Marking criteria

Your submission will be marked out of 100, and marks will be awarded as follows:

- *Program operates correctly (60 marks):*  
Criteria include: properly loading the dataset, correct operation of the methods in the class `EntryCatalog` (tested using a battery of JUnit tests), and an interactive and well-balanced GUI operating as expected providing the correct answers to use (including pie chart, the statistics and the GUI controls).  
Marks will be split into 30 marks for data loading and `EntryCatalog` class, and 30 marks for the GUI.
- *Use of Java features and quality of implementation (20 marks):*  
Criteria include aspects such as: use of inheritance (e.g., for the main classes extending the abstract classes provided), use of Java collections methods to handle collections correctly, use of Java 2D, use of Java Swing and event handling, and choice of programming structures for loops and conditionals.
- *Good programming style (20 marks):*  
Criteria include aspects such as: economy of code (i.e. methods are of reasonable length), class cohesion (i.e. classes have a clear set of responsibilities), readability (i.e. no long lines of code), sensible choice for class and variable names, use of comments in each class, and Javadoc documentation.

**Note: You have been provided with a project that compiles and executes correctly. If the project in your submission commit is broken (does not compile or execute), you will receive zero marks for this assignment.**

## Questions

If you have any questions about the assignment, please have another look at the template code and the handout provided to you. If your question is not answered there, please use the COM114 Discussion board set up for this Assignment. The lecturers and the demonstrators will be available to answer questions about the assignment in the practicals.

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November 2025